MorphoCLIP: Text-Supervised Contrastive Learning for Perturbation Matching in Cell Painting Images

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***Abstract*—Cell Painting microscopy produces multi-channel phenotypic images of cells under chemical or genetic perturbation, but matching those images back to the underlying perturbation is difficult and remains a bottleneck for phenotypic drug discovery. We present MorphoCLIP, a contrastive learning framework that embeds Cell Painting images and natural-language descriptions of perturbations in a shared 512-dimensional space. The image branch passes each of five fluorescence channels through a frozen DINOv3 ViT-L/16 backbone and aggregates the resulting CLS tokens with a single-layer CrossChannelFormer. The text branch uses a frozen BioClinical ModernBERT with a trainable projection head. Training uses the Continuously Weighted Contrastive Loss (CWCL) to handle soft-positive relationships across perturbations that share a biological target. On the CPJUMP1 benchmark (51 plates, 817 candidate perturbations), MorphoCLIP reaches text-to-image Recall@10 of 24.3% with a median rank of 28, around two orders of magnitude above the random-retrieval baseline of 0.12%. Unlike prior contrastive methods restricted to small molecules, MorphoCLIP is trained jointly on compounds, CRISPR knockouts, and ORF overexpressions in a single model, and supports both retrieval directions at inference.**

***Index Terms*—contrastive learning, cell painting, morphological profiling, CLIP, perturbation matching**

I. Introduction

Phenotypic drug discovery asks a deceptively simple question: what does this molecule, knockout, or overexpression construct actually do to a cell? Answering it well at scale could shorten the decade-long, $\sim$\$2 B path from candidate to approved drug [1], where roughly nine out of ten compounds entering clinical trials still fail [2]. Most of the existing answers come from expensive target-specific assays.

High-content microscopy offers a more scalable alternative. The Cell Painting assay [3], [4] stains live cells with five fluorescent dyes covering DNA, endoplasmic reticulum, RNA/nucleoli, actin/Golgi, and mitochondria, producing a multi-channel image of cellular state. Morphological changes after perturbation encode the perturbation's mechanism. The CPJUMP1 benchmark [5] casts this as a retrieval problem: given $\sim$3 million field-of-view images covering 303 compounds, 160 CRISPR knockouts, and 176 ORF overexpressions, can a model match perturbations that act on the same biological target?

Existing methods make uneven progress on this question. The classical CellProfiler pipeline [6] extracts handcrafted shape, texture, and intensity features and recovers only 5–25% of expected compound–gene matches on CPJUMP1 [5]. Recent contrastive methods do better but each leaves something on the table. CLOOME [7] pairs cell images with Morgan fingerprints and reaches 70$\times$ random retrieval, but only for compounds. MolPhenix [8] pushes this further with a pre-trained Phenom-1 encoder, again restricted to compounds. CellCLIP [9] replaces fingerprints with text prompts and a CrossChannelFormer over DINOv2-g (1.48B parameters), enabling all three perturbation types via language, but its cross-class mAP still trails CellProfiler. CWA-MSN [10] attacks the problem from another angle, using cross-well alignment for batch-effect correction; with only 22M parameters it beats CellCLIP by 9% on gene–gene retrieval, but it has no text branch and so does not support natural-language querying.

We propose **MorphoCLIP**, a contrastive learning framework that combines three ingredients no prior method has unified: text supervision, cross-well alignment for batch correction, and joint training over both small molecules and genetic perturbations. Following CLIP [11], MorphoCLIP aligns Cell Painting images and natural-language perturbation descriptions in a shared 512-dimensional space. The working hypothesis is that text supervision is complementary to batch correction: it adds semantic structure that purely visual methods cannot recover, while CWA neutralizes the technical variation that swamps fine-grained biological signal.

The contributions of this paper are:

1) A contrastive learning framework that combines text supervision, cross-well alignment, and joint compound/gene training, none of which prior methods have combined.
2) A parameter-efficient design built on a frozen DINOv3 ViT-L/16 backbone [12] ($\sim$300M frozen) and a CrossChannelFormer aggregator, with only $\sim$8M trainable parameters versus CellCLIP's 1.48B total.
3) Joint training on compounds, CRISPR knockouts, and ORF overexpressions in a single model, where prior contrastive methods are limited to compounds.
4) An evaluation on CPJUMP1 reaching text-to-image Recall@10 of 24.3% with a median rank of 28 out of 817 candidates.

II. Problem Statement

Given a Cell Painting image $\mathbf{x} \in \mathbb{R}^{5\times H\times W}$ (five fluorescence channels at 1080×1080 pixels) and a textual description t of the perturbation applied to the imaged cells, we learn encoders f_{img} and f_{txt} that map both modalities into a shared unit sphere:

$$f_{\text{img}}(\mathbf{x}),\ f_{\text{txt}}(t) \in \mathbb{R}^d, \quad \|f_{\text{img}}(\mathbf{x})\|_2 = \|f_{\text{txt}}(t)\|_2 = 1. \quad (1)$$

Matching image–text pairs should have high cosine similarity and non-matching pairs low similarity. We use $d = 512$.

The trained encoders support three retrieval tasks:

- **Text-to-image retrieval.** Given a perturbation description (e.g., "a compound that inhibits tubulin and disrupts cell division"), retrieve cell images that exhibit the corresponding phenotype.
- **Image-to-text retrieval.** Given a cell image, return the perturbation description that best matches the observed morphology.
- **Image-to-image retrieval.** Compare compound images against genetic perturbation images by cosine similarity to identify shared targets, the standard CPJUMP1 task. The text encoder shapes the embedding space during training but is not used at inference for this task.

Three properties of the data make the problem harder than standard vision–language alignment.

a) Batch effects.: The same perturbation produces visually different images across plates, days, and microscopes. Illumination drift, staining inconsistencies, and focus variation introduce technical variance that often dominates the biological signal. In CPJUMP1, replicability varies by modality (compounds replicate more cleanly than CRISPR) and by cell line (U2OS vs. A549) [5]. The mAP distribution is heavily right-skewed: most perturbations score close to zero.

b) Fuzzy ground truth.: Biological perturbations do not admit clean positive/negative labels in the way natural image–caption pairs do. Two compounds may share a primary target but diverge on secondary effects; a CRISPR knockout and a small molecule can converge on the same pathway through different mechanisms. Labels live on a continuum of similarity, which calls for soft-positive loss formulations rather than InfoNCE-style hard negatives.

c) Scale and compute.: CPJUMP1 contains 51 plates and 3.31 TiB of raw 16-bit TIFF images, covering $\sim$3 million field-of-view images of $\sim$75 million cells. End-to-end training of full vision and language encoders at this scale is not feasible on a single consumer GPU. We address this with a pre-extraction and caching pipeline that decouples backbone forward passes from the contrastive training loop.

III. Related Work

A. Morphological Profiling

Cell Painting [3] introduced the now-standard five-stain protocol for high-content morphological profiling and has become the dominant assay for high-throughput phenotypic screening [4]. The classical analysis tool, CellProfiler [6], extracts more than 1,000 handcrafted shape, texture, and intensity features per cell. On CPJUMP1 [5], this pipeline retrieves 5–25% of expected sister perturbations within a modality and 7–17% of CRISPR guide matches; cross-modal compound-to-gene matching is harder still.

Handcrafted features are bounded by the measurements their authors chose to compute and miss higher-order phenotypic patterns. Self-supervised vision transformers trained with DINO surpass CellProfiler on drug-target and gene-family classification with zero-shot transfer [13], which motivates the use of frozen foundation-model features for Cell Painting.

B. Contrastive Learning for Cell Painting

CLOOME [7] was the first CLIP-style framework for Cell Painting. It jointly trains a ResNet image encoder and a Morgan-fingerprint MLP and reaches top-1 retrieval $> 70\times$ random over $\sim$2,000 candidates. CLOOME handles only chemical perturbations and treats the five channels as RGB, discarding channel-specific signal.

MolPhenix [8] replaces the trained image branch with a frozen Phenom-1 encoder, averages replicate embeddings, and introduces a soft-to-label (S2L) loss, gaining $8.78\times$ over CLOOME. It also covers compounds only and depends on a proprietary backbone.

CellCLIP [9] is the closest prior work. It contributes (i) templated text prompts ("A Cell Painting image of [cell_type] cells treated with [drug_name], SMILES: [SMILES_string]"), (ii) a CrossChannelFormer that processes each channel separately before cross-attention, (iii) attention-based MIL pooling across cells, and (iv) the Continuously Weighted Contrastive Loss (CWCL) for soft positives. CellCLIP runs on a DINOv2-g backbone (1.1B parameters, 1.48B total) and is evaluated on the Bray dataset [14] and on RxRx3-core gene–gene benchmarks (CORUM recall 0.354). Although CellCLIP supports all perturbation types through text, its cross-class mAP falls below the CellProfiler baseline, suggesting that text supervision on its own does not solve the batch-effect problem.

C. Batch-Effect Correction

CWA-MSN [10] attacks batch effects directly with cross-well alignment inside a masked siamese network. By aligning embeddings of cells subjected to the same perturbation across different wells and plates, it absorbs technical variance without an explicit batch label. With 22M parameters and 0.2M training images, CWA-MSN improves on CellCLIP by 9% and on OpenPhenom by 29% on gene–gene retrieval (CORUM recall 0.386). It has no text branch.

D. Foundation Models for Microscopy

DINOv2 [15] and DINOv3 [12] provide self-supervised visual representations that transfer well to biological imaging despite being trained on natural images [13]. DINOv3 adds gram anchoring and an improved training recipe over DINOv2. Channel Vision Transformers [16] show that per-channel processing followed by cross-channel attention is

TABLE I
LANDSCAPE OF CELL PAINTING REPRESENTATION LEARNING METHODS. MORPHOCLIP UNIQUELY COMBINES TEXT SUPERVISION, GENE-INCLUSIVE TRAINING, AND BATCH CORRECTION.

Method	Image Encoder	Pert. Types	Batch Corr.	Params
CellProfiler [6]	Handcrafted	All	Sphering	N/A
CLOOME [7]	ResNet	Compounds	No	$\sim$25M
MolPhenix [8]	Phenom-1	Compounds	No	$\sim$36M
CellCLIP [9]	DINOv2-g	All (text)	No	1,477M
CWA-MSN [10]	ViT-S/16	Unimodal	CWA	22M
MorphoCLIP	DINOv3-L	All (text)	CWA	$\sim$8M

effective for multi-channel microscopy, which motivates our CrossChannelFormer aggregator.

On the text side, BioClinical ModernBERT [17] (150M parameters) is pre-trained on 53.5B tokens of PubMed, PMC, and clinical text, roughly 17$\times$ the biomedical corpus used by PubMedBERT (110M, 3.1B tokens) in CellCLIP. Built on ModernBERT with RoPE, Flash Attention, and GeGLU activations, it can encode both small-molecule and gene descriptions in a single model, unlike SMILES-specialized encoders such as ChemBERTa.

E. Positioning of MorphoCLIP

Table I places MorphoCLIP relative to these methods. The gap we target is the absence of any prior method that combines text supervision with both gene-inclusive training and batch-effect correction. CellCLIP has text but no gene training and no batch correction; CWA-MSN has genes and batch correction but no text. MorphoCLIP fills that intersection with a parameter-efficient architecture ($\sim$8M trainable parameters versus CellCLIP's 1.48B total).

IV. PROPOSED APPROACH

MorphoCLIP has two encoding branches and one contrastive objective. The image branch turns a five-channel Cell Painting field of view into a 512-dimensional unit vector; the text branch does the same for a natural-language prompt describing the perturbation. Figure 1 shows the full pipeline.

A. Image Encoder

Each Cell Painting site has five fluorescence channels (Table II). We process each channel independently: the single-channel image is replicated to pseudo-RGB at 384×384 resolution and passed through a frozen DINOv3 ViT-L/16 [12] (300M parameters), producing five 1024-dimensional CLS tokens per site,

$$\{\mathbf{z}_c\}_{c=1}^{5}, \quad \mathbf{z}_c \in \mathbb{R}^{1024}. \tag{2}$$

We use DINOv3 ViT-L rather than the DINOv2-g backbone of CellCLIP. The two have comparable feature quality on transfer benchmarks, but DINOv3 ViT-L is roughly 4$\times$ smaller, which lets us pre-extract a full plate in $\sim$7 minutes on a single RTX 5080. Features are written to disk as `.pt` files ($\sim$3 GB per plate) and reused across every training run, removing the need to re-run the backbone each epoch. A

TABLE II
CELL PAINTING FLUORESCENCE CHANNELS USED IN MORPHOCLIP.

Ch	Stain	Target
1	MitoTracker / Alexa 647	Mitochondria
2	Phalloidin / Alexa 568	Actin
3	WGA / Alexa 488	Golgi / plasma membrane
4	Concanavalin A / Alexa 488	Endoplasmic reticulum
5	Hoechst 33342	DNA / nucleus

sanity check on the cached features shows that the per-channel encoding preserves channel-specific signal: mean intra-site cross-channel cosine similarity is 0.82, and the DNA channel is the most distinct from the others (mean similarity 0.73).

a) CrossChannelFormer (CCF).: To collapse the five channel tokens into a single image vector, we use a single-layer transformer encoder [18] with 4 attention heads, following the channel-aware processing of CellCLIP [9] and Channel ViT [16]. The CCF lets the model attend across channels so it can pick up cross-channel co-occurrences, for example a joint signal in mitochondrial and ER channels. The transformer outputs are mean-pooled across the channel dimension:

$$\mathbf{h}_{\text{img}} = \text{MeanPool}(\text{CCF}(\mathbf{z}_1, \mathbf{z}_2, \mathbf{z}_3, \mathbf{z}_4, \mathbf{z}_5)). \tag{3}$$

We also report a baseline that drops the CCF and averages the five channel tokens directly.

b) Image projection head.: A two-layer MLP with LayerNorm, GELU, and dropout ($p = 0.3$) maps $\mathbf{h}_{\text{img}}$ into the shared 512-dimensional space:

$$f_{\text{img}}(\mathbf{x}) = \text{L2Norm}\big(\mathbf{W}_2 \cdot \text{Drop}(\text{GELU}(\text{LN}(\mathbf{W}_1 \cdot \mathbf{h}_{\text{img}})))\big), \tag{4}$$

with $\mathbf{W}_1 \in \mathbb{R}^{512\times1024}$ and $\mathbf{W}_2 \in \mathbb{R}^{512\times512}$.

B. Text Encoder

We convert the perturbation metadata into a verbose prompt using one of four modality-specific templates, populated from external biological databases:

- **Compound:** *"Chemical perturbation: {name}. Target: {target_protein} ({gene_symbol}). Function: {mechanism_of_action}. SMILES: {canonical_smiles}."*
- **CRISPR:** *"CRISPR knockout of {gene_symbol}. Protein: {protein_name}. Function: {protein_function}. GO terms: {biological_process_terms}."*
- **ORF:** *"Overexpression of {gene_symbol}. Protein: {protein_name}. Function: {protein_function}."*
- **Negative control:** *"Negative control treatment (DMSO vehicle) applied to Cell Painting images."*

When an annotation field is missing, the prompt collapses to a perturbation-name-only description. Compound annotations come from ChEMBL and the CPJUMP1 metadata; gene annotations come from UniProt and Gene Ontology.

We encode each prompt with a frozen BioClinical ModernBERT [17] (150M parameters, 768-d CLS token), pre-trained on 53.5B tokens of PubMed, PMC, and clinical text.

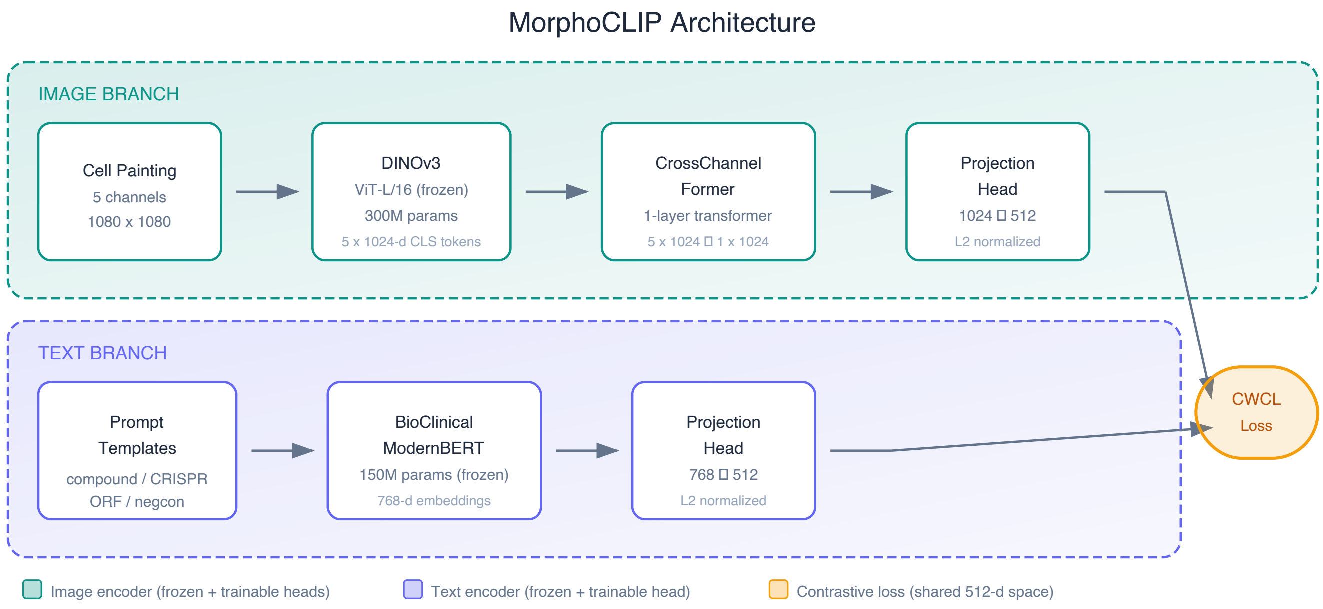

Fig. 1. MorphoCLIP architecture. A five-channel Cell Painting image (1080 × 1080) is passed channel-by-channel through a frozen DINOv3 ViT-L/16, producing five 1024-d CLS tokens. A single-layer CrossChannelFormer (4 heads) mixes the channel tokens, and the result is projected to the shared 512-d space. The text prompt is encoded by frozen BioClinical ModernBERT (150M parameters) and projected to the same space. CWCL pulls matching pairs together and pushes non-matches apart. Only the shaded components (CrossChannelFormer and the two projection heads) are trained, totalling ∼8M parameters.

A trainable projection head matches the image head and maps the CLS token to the 512-d space:

$$f_{\text{txt}}(t) = \text{L2Norm}\big(\mathbf{W}_4 \cdot \text{Drop}(\text{GELU}(\text{LN}(\mathbf{W}_3 \cdot \mathbf{h}_{\text{txt}})))\big), \quad (5)$$

with $\mathbf{W}_3 \in \mathbb{R}^{512\times768}$ and $\mathbf{W}_4 \in \mathbb{R}^{512\times512}$. The 768-d BERT features are pre-computed and cached separately from the projected 512-d outputs, which removes the BERT forward pass from the training loop and gives roughly an order-of-magnitude speedup per epoch.

C. Training Objective

a) Continuously Weighted Contrastive Loss (CWCL).: Standard InfoNCE [19] treats every off-diagonal pair as a hard negative, which is the wrong assumption when two compounds share a target. Following CellCLIP [9] and CWA-MSN [10], we use CWCL with continuous similarity weights: same `broad_sample` → 1.0, same target gene but different compound → 0.5–0.8, otherwise → 0.0. For a batch of N image–text pairs with logits $\mathbf{S} = \tau \cdot \mathbf{F}_{\text{img}} \mathbf{F}_{\text{txt}}^{\top}$ and weight matrix $\mathbf{W} \in [0,1]^{N\times N}$, the per-pair term is

$$\ell_i^{\text{img}\to\text{txt}} = -\sum_{j=1}^{N} \tilde{w}_{ij} \log \frac{\exp(S_{ij})}{\sum_{k=1}^{N} \exp(S_{ik})}, \quad (6)$$

where $\tilde{w}_{ij} = w_{ij} / \sum_k w_{ik}$ normalizes weights row-wise. The full loss is the symmetric average,

$$\mathcal{L}_{\text{CWCL}} = \frac{1}{2N} \sum_{i=1}^{N} \big(\ell_i^{\text{img}\to\text{txt}} + \ell_i^{\text{txt}\to\text{img}}\big). \quad (7)$$

The temperature τ is a learnable parameter (LogitScale module), initialized to $\ln(1/0.07)$ and clamped at $\ln(100)$.

b) Cross-Well Alignment (CWA).: Batch effects are the dominant confound in Cell Painting. CWA mitigates them by grouping embeddings by plate within each batch, subtracting the per-plate mean, and re-normalizing to the unit sphere:

$$\tilde{\mathbf{f}}_w = \frac{\mathbf{f}_w - \bar{\mathbf{f}}_{p(w)}}{\|\mathbf{f}_w - \bar{\mathbf{f}}_{p(w)}\|_2}, \quad \bar{\mathbf{f}}_p = \frac{1}{|W_p|} \sum_{w\in W_p} \mathbf{f}_w, \quad (8)$$

where $p(w)$ is the plate of well w and W_p is the set of wells from plate p in the current batch. CWA is applied to image embeddings only and requires a batch sampler that draws from multiple plates.

D. Optimization

We train with AdamW [20] at learning rate 10^{-4}, weight decay 0.2, $(\beta_1, \beta_2) = (0.9, 0.999)$, and $\varepsilon = 10^{-8}$. The schedule is a 100-step linear warmup followed by cosine decay over 100 epochs. We use FP16 mixed precision and gradient clipping at norm 1.0. The batch size is 512 wells, chosen to give the contrastive objective a useful pool of in-batch negatives. The trainable parameter count is ∼8M (CCF plus the two projection heads), so the optimizer state fits comfortably in the 16 GB of an RTX 5080. With pre-extracted features, an epoch takes 3–5 minutes wall-clock.

V. Experimental Setup

A. Dataset

We evaluate on the CPJUMP1 pilot dataset [5], a public benchmark from the JUMP Cell Painting Consortium hosted on the Cell Painting Gallery (AWS S3, prefix `cpg0000-jump-pilot`). CPJUMP1 contains 51 plates and 3.31 TiB of raw 16-bit TIFF images. The collection covers three perturbation types (303 small-molecule compounds, 160 CRISPR gene knockouts, 176 ORF overexpressions), two cell lines (U2OS and A549), and several timepoints. Each gene target is linked to at least two compounds, which allows systematic evaluation of compound–gene matching. Each plate has 384 wells with 7–16 sites per well; each site produces five fluorescence channels at 1080×1080 pixels plus three brightfield channels that we do not use.

B. Data Pipeline

We use a stream-extract-delete pipeline to keep disk usage bounded. Raw TIFFs ($\sim$58–118 GiB per plate) are downloaded to scratch space, passed through the frozen DINOv3 backbone, and removed once the per-channel CLS features are written. This compresses 3.31 TiB of raw data into roughly 153 GB of cached features ($\sim$3 GB per plate). Extraction takes about 7 minutes per plate on an RTX 5080, or roughly 6 hours for all 51 plates. Text embeddings are pre-computed in the same way: each perturbation prompt is encoded by frozen BioClinical ModernBERT and the 768-d CLS tokens are cached to disk.

The unit of training is a well with all of its sites aggregated. The cached feature tensor for a well has shape $(S, 5, 1024)$, where S is the number of sites. We mean-pool over the site dimension before the CrossChannelFormer to obtain a single $(5, 1024)$ tensor per well.

C. Train/Test Split

We implement three split strategies and evaluate the model under each:

- **Official representation** (default): CRISPR and ORF perturbations form the training set; compounds are split by timepoint into validation and test.
- **Gene–compound holdout**: a 60/20/20 split at the target-gene level, so no target gene appears in both train and test.
- **CellCLIP-style**: a 75/25 split within perturbation slices, matching CellCLIP's evaluation protocol.

All three splits keep wells from the same perturbation together; no well is split across partitions.

D. Evaluation Protocol

We report numbers on a held-out validation split of 2,220 wells spanning 817 candidate perturbations. Both retrieval directions are reported:

- **Recall@k** for $k \in \{1, 5, 10\}$: the fraction of queries whose correct match appears in the top-k retrieved candidates.
- **Median rank**: the median position of the correct match in the full ranked list (lower is better).

Random retrieval over 817 candidates gives Recall@1 $\approx$ 0.12% and median rank $\approx$ 408, which we use as the reference baseline throughout.

TABLE III
Factorial ablation design.

Variant	CWA	Genes	Text	Tests
Full MorphoCLIP	✓	✓	✓	Complete method
No text	✓	✓		CWA-MSN–like
No genes	✓		✓	CellCLIP + CWA
No CWA		✓	✓	Batch corr. effect

E. Baselines

We compare against three baselines. Each was reported on a different dataset or task formulation, so the comparisons below are indicative rather than strictly controlled, and we flag the relevant differences in the results section.

- **CellProfiler** [6]. Handcrafted features on CPJUMP1 sister-perturbation matching, fraction retrieved 4.3–25.1% (mean $\sim$13%) [5].
- **CellCLIP** [9]. Contrastive model on the Bray dataset [14] with text-to-profile retrieval. Best Recall@10 7.4%; CORUM gene–gene recall 0.354.
- **CWA-MSN** [10]. Self-supervised siamese network with cross-well alignment. CORUM gene–gene recall 0.386 with 22M parameters.

F. Implementation Details

Batch size 512 wells. AdamW optimizer with lr $= 10^{-4}$, weight decay 0.2, $(\beta_1, \beta_2) = (0.9, 0.999)$, $\varepsilon = 10^{-8}$. Schedule: 100-step linear warmup followed by cosine decay over 100 epochs. FP16 mixed precision; gradient clipping at norm 1.0. CrossChannelFormer: 1 transformer layer, 4 attention heads. Projection heads: hidden width 512, dropout 0.3. Temperature initialized to $\ln(1/0.07)$ and clamped at $\ln(100)$. Seed 42 for the headline number; the planned ablation runs use three seeds (42, 1337, 2024) to give variance estimates.

G. Planned Ablations

We design a 2^3 factorial ablation over three binary factors (text supervision, CWA batch correction, and gene-inclusive training) to attribute performance to each component:

A second planned ablation varies prompt richness across four levels: (1) perturbation name only, (2) name plus target and function, (3) name plus target plus SMILES or GO terms, and (4) full description with pathway annotations. The goal is to measure how much of the gain comes from injecting structured domain knowledge into the prompt, and in particular whether SMILES strings carry useful signal through BioClinical ModernBERT, which was not pre-trained on chemical notation.

TABLE IV
CROSS-MODAL RETRIEVAL. MORPHOCLIP IS EVALUATED ON CPJUMP1 (2,220 WELLS, 817 CANDIDATE PERTURBATIONS). CELLCLIP NUMBERS ARE TAKEN FROM THE ORIGINAL PAPER, EVALUATED ON THE BRAY DATASET. CELLPROFILER USES PROFILE-BASED SISTER-PERTURBATION MATCHING ON CPJUMP1.

Model	Direction	R@1	R@5	R@10
CellProfiler	sister matching	FR: 4.3–25.1%		
CellCLIP	perturb → profile	1.18	4.49	7.37
	profile → perturb	1.25	4.82	7.39
MorphoCLIP	image → text	5.72	6.71	7.07
	text → image	3.51	**14.86**	**24.32**

TABLE V
WALL-CLOCK PROFILE OF THE TRAINING LOOP ON A SINGLE RTX 5080 (16 GB VRAM). THE LORA ROW IS AN INDICATIVE ESTIMATE FROM A PARTIAL RUN, INCLUDED TO MOTIVATE THE PRE-EXTRACTION DESIGN RATHER THAN AS A MEASURED BASELINE.

Configuration	Train. Params	VRAM	Time/epoch
Pre-extracted + CCF (ours)	$\sim$8M	$\sim$2 GB	3–5 min
Frozen DINOv3 + LoRA (r=8, est.)	$\sim$20M	$\sim$14 GB	45–90 min

VI. RESULTS AND DISCUSSION

A. Cross-Modal Retrieval Performance

Table IV reports retrieval performance of MorphoCLIP on the CPJUMP1 validation split alongside the published CellCLIP and CellProfiler numbers. The baselines were measured on different datasets and task formulations, so the cross-method comparisons are indicative; we discuss the caveats below.

The headline number is text-to-image Recall@10 of 24.3%, two orders of magnitude above the random baseline of 0.12% on the same 817-candidate pool, with a median rank of 28 (top 3.4%). Image-to-text retrieval is substantially harder: Recall@10 is 7.07% with a median rank of 289 (still below the random baseline of 408, but only by a factor of about 1.4×).

B. Retrieval Asymmetry

The two directions are not symmetric: text-to-image R@10 is 24.3% versus image-to-text R@10 of 7.1%, with median ranks of 28 and 289 respectively. Most of this gap is structural rather than algorithmic. The validation split has many wells per perturbation, so a text query has multiple valid image positives to retrieve, while an image query must identify a single correct label out of 817. CellCLIP avoids this asymmetry because it operates on aggregated profiles (one vector per perturbation) rather than individual well images, making image-to-text and text-to-image largely equivalent on its setup.

C. Comparison with Baselines

a) Versus CellCLIP.: On text-to-image retrieval, MorphoCLIP reaches R@10 = 24.3% on a candidate pool of 817 perturbations, against CellCLIP's best reported R@10 of 7.4% on its evaluation set. The two evaluations are not strictly comparable: the candidate pools differ in size and composition, and CellCLIP retrieves over aggregated profiles while MorphoCLIP retrieves over individual well images, which favors the text-to-image direction (see above). On the more directly comparable image-to-text direction, MorphoCLIP R@10 of 7.07% is essentially tied with CellCLIP's profile-to-perturb R@10 of 7.39%. The contribution we want to highlight is therefore not raw R@10 but parameter efficiency: $\sim$8M trainable parameters versus CellCLIP's 1.48B total, with comparable image-to-text performance and joint training over all three perturbation types instead of compounds only.

b) Versus CellProfiler.: On CPJUMP1, CellProfiler's fraction retrieved ranges from 4.3% to 25.1% (mean $\sim$13%) depending on the modality and matching task [5]. MorphoCLIP's text-to-image R@10 sits above the upper end of that range, on the easier text-query direction. The qualitative difference, independent of any number, is that MorphoCLIP supports natural-language querying of an image database, which CellProfiler does not.

c) Versus CWA-MSN.: A direct head-to-head comparison with CWA-MSN requires running both methods on the same gene–gene retrieval benchmarks (CORUM, HuMAP, Reactome). CWA-MSN's CORUM recall of 0.386 with 22M parameters is the relevant target. The factorial ablation in Table III is designed to answer the more specific question of whether text supervision adds anything over CWA alone, which is the place we expect MorphoCLIP to provide a contribution.

D. Feature Analysis

We inspect the pre-extracted DINOv3 features to check that channel structure survives the frozen backbone. On plate BR00116991 (500 sites), a variance decomposition attributes 31.1% of feature variance to differences between channels, 37.1% to differences between sites, and 31.8% to residual variation. The between-channel component is large enough to suggest that the per-channel encoding does carry channel-specific signal, even though DINOv3 was trained on natural images. A PCA over the 5×1024-dimensional feature space reaches 90% of variance with 23 principal components, indicating moderate intrinsic dimensionality.

Cross-site same-channel similarity (mean 0.84) is higher than cross-channel similarity at the same site (mean 0.75), a gap of 0.09 that holds for every channel pair. The DNA/nucleus channel is the most distinct, with a mean similarity of 0.73 to the other four channels, consistent with its very different biological content.

E. Computational Efficiency

Pre-extraction is what makes MorphoCLIP cheap to iterate on (Table V). The DINOv3 forward pass is the dominant cost; once features are cached, the contrastive training loop is small enough to fit comfortably on a single 16 GB GPU.

Feature extraction is a one-time cost of about 7 minutes per plate ($\sim$6 hours for all 51). After that, an epoch over the cached features takes 3–5 minutes wall-clock, which is what enables the ablation study to run dozens of configurations on a single GPU in a day. Both DINOv3 (300M parameters) and BioClinical ModernBERT (150M parameters) stay frozen throughout; only the CrossChannelFormer and the two projection heads are updated.

VII. Conclusion

We have presented MorphoCLIP, a contrastive learning framework that maps Cell Painting images and natural-language perturbation descriptions into a shared 512-dimensional space. The design combines three ingredients that prior work has only paired in twos: text supervision, cross-well batch alignment, and joint training over both small molecules and genetic perturbations. The model trains on a single 16 GB GPU thanks to a pre-extraction pipeline that caches the frozen-backbone outputs.

The main empirical results are a text-to-image Recall@10 of 24.3% on 817 CPJUMP1 candidates with a median rank of 28, achieved with $\sim$8M trainable parameters (versus 1.48B total in CellCLIP), and image-to-text performance roughly on par with CellCLIP's profile-based retrieval despite the much smaller trainable footprint. The same model handles compounds, CRISPR knockouts, and ORF overexpressions, which lets text act as a common perturbation representation across modalities.

If the planned ablations confirm that the gains hold up when the three components are isolated, the practical use cases this opens up include searching for existing drugs whose cellular signature matches a target phenotype, predicting a likely mechanism of action by an unknown compound's neighbors in embedding space, and querying image databases in plain English without engineering bespoke features.

A. Limitations and Future Work

The two largest gaps in the current paper are the absence of completed ablations and the use of a single seed for the headline number. The factorial ablation in Table III is designed to attribute performance to text supervision, CWA, and gene-inclusive training individually; until those runs land, the contribution of each component is asserted rather than measured. The follow-up multi-scale text ablation will say whether SMILES strings and GO-term injections actually add signal through BioClinical ModernBERT or are decorative.

Beyond ablation completion, three directions are worth pursuing. First, scale: CPJUMP1 is the pilot, and the full JUMP corpus contains $\sim$140,000 perturbations across many labs, microscopes, and cell lines, which is the right setting to test generalization. Second, richer prompts: pulling pathway and mechanism annotations from UniProt, Reactome, and DrugBank should strengthen the text encoder's biological grounding, and the multi-scale ablation will quantify how much that helps. Third, zero-shot transfer: running the trained encoders on Cell Painting datasets from labs and cell lines not seen during training would tell us whether the learned representations are general or specific to CPJUMP1.

We plan to release model weights, the evaluation suite, and an interactive natural-language retrieval demo on publication.

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